

Narrative Contagion at the Sovereign Default Boundary*

[Author]

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Abstract

Stories about sovereign default spread from investor to investor, and sometimes they move spreads. I put a contagious pessimistic narrative into a sovereign debt-pricing model with a default boundary and a feedback from borrowing costs to fundamentals. The pricing side is a theorem from one primitive: because default probability is an S-curve in fundamentals, a story that shifts perceived fundamentals is priced in proportion to the curve’s local slope. The contagion side rests on one stated behavioral assumption, that stories spread when they are worth acting on, under which the contagion rate inherits the same hump. The state space splits into three derived regions: an immune region where the story dies regardless of seeding, a febrile zone where it goes viral and is priced but self-limits, and a critical zone where the narrative-spread-fundamentals loop tips the sovereign into a default that would not otherwise occur: the story does not move the sovereign to the cliff. It moves the cliff to the sovereign. In scenarios anchored to sovereign haircut and spread magnitudes, a story at ten percent prevalence is worth 229 basis points at the boundary and 2.3 off it, and a seeded outbreak adds at most 5.2 percentage points to the two-year default hazard at the baseline feedback, 0.7 to 6.9 across the gross-financing-need band, all inside the zone. Policy divides sharply: deleting three quarters of the story’s carriers buys about six weeks, transparency dampens, and a credible spread cap sterilizes the epidemic itself, at 17 percentage points of default probability per year of delay.

JEL codes: D83, D84, E71, F34, G01, H63.

Keywords: narratives, epidemiology, sovereign debt, default boundary, doom loop, text data.

1 Introduction

On July 26, 2012, the president of the European Central Bank told a London investment conference that the ECB was ready to do “whatever it takes to preserve the euro.” No facility existed. The Outright Monetary Transactions program the speech foreshadowed was announced

*First draft: July 2026. Preliminary and incomplete. Comments welcome. Estimates requiring text-data assembly are marked as pending throughout. Every number in the text is either documented in a cited source or an output of the model solved here. Replication code in Julia and R accompanies the draft.

only in September and has never bought a bond. Yet two-year Italian and Spanish yields fell by roughly two percentage points around the OMT announcements (Altavilla et al., 2016), with the declines concentrated in default and redenomination premia rather than expected rates (Krishnamurthy et al., 2018). Three sentences repriced two of the largest bond markets in the world. The same three sentences, spoken in 2006 about the same two sovereigns, would have moved nothing: there was, then, no story to kill.

This paper is about when words and the stories they carry can move sovereign spreads, and it insists on deriving the answer rather than assuming it. The narrative-economics program of Shiller (2017) models stories as epidemics: beliefs that pass from person to person with a reproduction rate and a recovery rate. Applied to sovereign debt, the program faces an immediate referee objection. If a pessimistic debt story is simply postulated to infect investors and move prices, the theory explains everything and predicts nothing: any spread movement not accounted for by fundamentals becomes “narrative.” The discipline this paper imposes is a boundary condition, and I am precise about its epistemic anatomy. Stories about default are priced, and spread, only where default is genuinely in play. The pricing half is a theorem from one primitive. The contagion half rests on one behavioral assumption, stated as such and given a numbered rival.

The primitive is the shape of the default probability. A sovereign defaults when fundamentals cross a threshold, so its one-year default probability is an S-curve in current fundamentals: flat near zero for safe sovereigns, flat near one for condemned ones, steep in between. A narrative, in this model, is a portable pessimistic belief: investors who carry the story price the bond as if fundamentals were worse by a fixed amount ξ : the books are cooked, the union will break. The story’s effect on the spread is then proportional to the probability the extra pessimism actually changes the default assessment, which is the S-curve’s local slope: a hump that peaks where default hangs in the balance and vanishes in both tails. That is the pricing half of the boundary condition, and it is a theorem, not a specification choice. The contagion half follows from asking why anyone repeats a story. Following the fitness logic of Flynn and Sastry (2024), investors adopt a story when it is worth acting on, and a story about default is worth acting on exactly when it moves valuations: the adoption rate, and hence the epidemic reproduction number $R_0(\theta)$, inherits the same hump. That step is Assumption 1, a behavioral postulate I state rather than derive, with its rival kept in play until the data can separate them. One measurable object, the local sensitivity of spreads to fundamentals news, governs where stories are priced, where they spread, and, through the feedback from spreads to debt service, where they can kill.

The theory delivers three regions. Where $R_0 < 1$, seeding a story produces a fizzle: prevalence decays geometrically and the pricing effect never exceeds the seed times the hump, which for safe sovereigns is a few basis points. Where $R_0 > 1$, a seed becomes an outbreak: prevalence converges to an endemic level and spreads carry a narrative premium for as long as fundamentals stay in the zone: viral, priced, but self-limiting. And in an inner critical region, the loop closes: the outbreak raises spreads, spreads drain fundamentals through debt service, worse fundamentals raise both the story’s price impact and its contagion rate, and the sovereign is dragged across

a boundary it would not otherwise have crossed. The geometry deserves emphasis, because it inverts the usual metaphor: the deterministic basin of default is bounded by a curve that shifts with prevalence, so the story does not push the sovereign toward a fixed cliff. It moves the cliff. In the calibrated model, a sovereign at spread 465 basis points survives indefinitely without the story and, seeded with a two percent prevalence, is defaulted at month 42: a slow narrative crisis, in the sense of Lorenzoni and Werning (2019), whose engine is epidemiology rather than equilibrium selection.

I want to be precise about what the quantitative claims are, because the honest ones are asymmetric. Narratives in this model move prices much more than they move defaults. At the hump’s peak, ten percentage points of prevalence are worth 229 basis points of spread, and a full outbreak generates an excess spread that reaches 3198 basis points before the end: magnitudes in the range of the fundamentals-unexplained spread components documented for the euro periphery by De Grauwe and Ji (2013). The same outbreak adds at most 5.2 percentage points to the two-year default hazard at the baseline feedback, a band of 0.7 to 6.9 across the range of gross financing needs that disciplines the feedback parameter, and essentially zero outside the zone: for a safe sovereign the addition is 0.06 percentage points. Tweets do not bankrupt safe sovereigns, in this model or, I suspect, anywhere. What a story can do is tax a vulnerable sovereign heavily while it circulates, and occasionally, in a narrow band of fundamentals, finish one off.

The policy analysis divides interventions by what they do to the epidemic, and the division is sharp enough to be the paper’s practical contribution. Counter-speech that deletes carriers, takedowns and official denials, fails against a supercritical story for the reason quarantine-evasion fails against any epidemic: prevalence regrows at rate $\gamma(R_0 - 1)$, so removing three quarters of carriers postpones the outbreak by $\ln 4 / [\gamma(R_0 - 1)]$, about six weeks at the calibration’s zone center. Transparency, modeled as a faster abandonment rate for stories contradicted by verifiable data, is genuine prophylaxis: it shrinks the susceptible zone from both ends and cuts the hazard wedge from 5.2 to 3.0 percentage points. And a credible spread cap, the whatever-it-takes instrument, does something neither can: by breaking the link from prevalence to price, it removes the story’s decision-relevance, which starves the contagion itself: the wedge falls to 0.0. The cap does not censor the story: it makes the story pointless. Its cost is time-sensitive, however, because the feedback loop runs while policy deliberates: at the calibration, each year of delay in introducing the cap costs 17 percentage points of default probability, 4 to 41 across the financing-need band. July 2012 arrived roughly thirty months after Greek spreads first signaled the zone. The model prices that interval.

The paper works a well-defined seam between literatures. From narrative economics it takes the epidemiological structure (Shiller, 2017; Kermack and McKendrick, 1927) and the fitness microfoundation (Flynn and Sastry, 2024), and gives the program what survey evidence says it needs, a setting where narratives are decision-relevant (Andre et al., 2026): here they are priced, and their prevalence and reproduction rate are estimable from text. From sovereign-debt theory it takes the default boundary and the service feedback (Eaton and Gersovitz, 1981; Arellano, 2008; Calvo, 1988; Lorenzoni and Werning, 2019), and offers a mechanism for belief-driven crises

that requires no equilibrium multiplicity: unlike the sunspot coordination of Cole and Kehoe (2000) and Aguiar et al. (2022), the narrative equilibrium is unique period by period, and the belief object is a measurable epidemic rather than an unobservable selection device: closer in spirit to the sentiments of Angeletos and La’O (2013), but with dynamics and a data counterpart. From the text-as-data literature (Gentzkow et al., 2019; Baker et al., 2016; Bybee et al., 2020) it takes measurement tools, and returns a theory of *which* text objects should price: prevalence and reproduction of specific default narratives, not sentiment at large, with the boundary interaction supplying the identification that predictive regressions of the Dergiades et al. (2015) kind lack.

Two boundaries on the claims, stated at the outset. The behavioral core, that investors adopt stories in proportion to their observed price relevance, is an assumption about social learning, not a theorem. Section 2 states it explicitly, and Section 6 shows it is testable against its natural alternative, adoption driven by perceived mispricing, because the two disagree about whether narrative volume survives a credible cap. And causality in the empirical designs runs uphill against the model’s own prediction that narratives respond to fundamentals: the designs therefore lean on scheduled-release timing, cross-border story spillovers, and the theory’s built-in placebo, outbreaks in immune countries, rather than on any pretense that text is exogenous.

My plan is as follows. Section 2 sets out the pricing and contagion blocks. Section 3 derives the hump, the zone, and the three regions. Section 4 analyzes deletion, transparency, and the cap. Section 5 reports calibrated magnitudes. Section 6 specifies measurement and empirical designs. Section 7 concludes. Proofs and extensions are in the appendix.

2 A Sovereign, Its Creditors, and a Story

The results divide into three epistemic classes, and I mark them as they arrive. The pricing results, Lemma 1 and Proposition 1, are theorems from the threshold-default primitive alone. The contagion and policy results, Propositions 2 through 4, are theorems conditional on Assumption 1, which is behavioral and has a stated rival. Every magnitude in Section 5 is scenario arithmetic: the epidemic block has no estimated parameters yet, and the text says so wherever they bind.

2.1 Fundamentals, default, and pricing

Time is monthly, and pricing horizons are annual. A sovereign’s fundamentals θ_t measure its debt-service capacity: the distance, in units of annual resources, between what the government can mobilize and what its obligations require. Fundamentals mean-revert and are drained by borrowing costs:

$$\theta_{t+1} = \rho_m \theta_t + (1 - \rho_m) \bar{\theta} - \chi_m s_t + \varepsilon_{t+1}, \quad \varepsilon \sim N(0, \sigma_m^2), \quad (1)$$

where s_t is the spread on the sovereign’s debt and χ_m converts a year of elevated refinancing costs into lost capacity: the debt-service feedback of Calvo (1988) and Lorenzoni and Werning (2019). Default occurs when capacity is exhausted, $\theta_t < 0$. At annual horizon, with annualized

parameters (ρ, σ, χ) , the default probability perceived by an investor who believes fundamentals are θ is

$$p(\theta, s) = \Phi(z(\theta, s)), \quad z(\theta, s) = \frac{-\rho\theta - (1 - \rho)\bar{\theta} + \chi s}{\sigma}. \quad (2)$$

The S-shape of p in θ is the paper's load-bearing object, and it comes free: any default technology in which a shock-buffed state crosses a threshold produces one.

A pessimistic narrative circulates among the sovereign's creditors. Carrying the story means pricing the bond under fundamentals worse by ξ : the deficit is understated, the peg will break. The story is portable and self-contained, which is what lets it spread. Its content is a belief distortion, not information. With prevalence n_t , the fraction of investor wealth carrying the story, the market-clearing default assessment is the wealth-weighted average

$$\bar{p}(\theta, s, n) = (1 - n) \Phi(z) + n \Phi(z + \Delta), \quad \Delta \equiv \frac{\rho\xi}{\sigma}, \quad (3)$$

so the story enters as a shift of size Δ in the standardized distance to default for the infected. Risk-neutral pricing of one-period debt with recovery $\mathcal{R}$ gives the spread as expected loss, $s = (1 - \mathcal{R})\bar{p}$, and since the spread feeds back into the default assessment through (2), the within-period equilibrium is a fixed point.

Lemma 1 (Pricing equilibrium and the doom-loop multiplier). *If $(1 - \mathcal{R})\chi/(\sigma\sqrt{2\pi}) < 1$, the map $s \mapsto (1 - \mathcal{R})\bar{p}(\theta, s, n)$ is a contraction: for every (θ, n) there is a unique equilibrium spread $s^*(\theta, n)$, increasing in n and decreasing in θ . Small shocks to the default assessment are amplified by the multiplier*

$$M(\theta, n) = \left[1 - (1 - \mathcal{R}) \frac{\chi}{\sigma} ((1 - n)\phi(z^*) + n\phi(z^* + \Delta)) \right]^{-1} \geq 1,$$

which peaks at the boundary and equals 2.8 at its calibrated maximum. If the condition fails, an interval of fundamentals admits multiple pricing equilibria: the Calvo (1988) regime, mapped in Appendix B.

The calibration sits inside the uniqueness region with a maximal loop gain of 0.64, and I keep it there deliberately. Everything fragile in this paper happens in a unique equilibrium: the narrative moves a well-defined price, not a selection among prices, which is what makes its prevalence a regressor rather than a sunspot.

2.2 The contagion block

Prevalence evolves by the logic of Kermack and McKendrick (1927): carriers meet susceptibles and transmit, and carriers abandon the story at rate γ :

$$n_{t+1} - n_t = \beta(\theta_t, n_t) n_t (1 - n_t) - \gamma n_t. \quad (4)$$

The economics is in β , and here the paper departs from specification toward microfoundation. A story is transmitted when a susceptible investor who hears it decides it is worth adopting, and

an investor whose business is prices adopts a story in proportion to what it does to prices.

Assumption 1 (Decision-relevant adoption). *Upon meeting a carrier, a susceptible investor adopts the story with probability $h(v)$, where $v = \partial s^* / \partial n$ is the story's observed price relevance, h is increasing, and $h(0) = 0$: a story that cannot move the price is not worth retelling. Meetings occur at rate $\bar{m}A$, where A indexes the amplification environment: the reach and velocity of the media that carry financial stories. Hence $\beta(\theta, n) = \bar{m}A h(v(\theta, n))$.*

The assumption is the fitness principle of Flynn and Sastry (2024) specialized to traders: narratives survive when they organize action, and in a bond market, action means repricing. It is behavioral, and I flag its main rival now because the two are separated by data later: adoption could instead track the story's *perceived mispricing*, the gap between the market price and the infected valuation. The two rules agree everywhere except under a price peg, where relevance dies and mispricing survives, and Section 4 turns that disagreement into a test. Under Assumption 1, the story's epidemiology is pinned to its pricing: the reproduction number

$$R_0(\theta) = \frac{\beta(\theta, 0)}{\gamma} = \frac{\bar{m}A h(v(\theta, 0))}{\gamma} \quad (5)$$

is large exactly where the story is worth money, and the question of where stories can go viral becomes the question of where beliefs are priced. That question has an exact answer, and it is the next section's business.

3 The Boundary Theorems

3.1 Where stories are priced

Proposition 1 (The pricing hump is derived, not assumed). *The story's equilibrium price impact is*

$$\frac{\partial s^*}{\partial n} = (1 - \mathcal{R}) [\Phi(z^* + \Delta) - \Phi(z^*)] M(\theta, n).$$

(i) *The belief gap $\Phi(z^* + \Delta) - \Phi(z^*)$ is the mass of the fundamentals density inside a window of width Δ : it is log-concave and single-peaked in z^* , maximal where the marginal default assessment is even, $\bar{p} \approx 1/2$, and vanishing in both tails. (ii) Since $z^*(\theta)$ is strictly decreasing, the same statement holds in θ : the price impact is single-peaked in fundamentals, converging to zero both for safe sovereigns ($\theta \rightarrow \infty$) and for condemned ones ($z^* \rightarrow \infty$). (iii) The multiplier $M \in [1, \bar{M}]$ inherits its peak from the same density and distorts the hump only within the bounded factor $\bar{M}$.*

This is the proposal's central specification assumption converted into a theorem: nothing about narratives was assumed except that they shift perceived fundamentals by a fixed amount, and the hump follows from the S-curve alone. The proposition also disposes of both polar objections at once. Stories cannot be priced into safe sovereigns because the extra pessimism changes no default assessment: at the calibrated safe point, ten percentage points of prevalence are worth 2.3 basis points. And stories add little to condemned sovereigns because default is

already the assessment: pessimism about a foregone conclusion is redundant. The action is in between, where ten points of prevalence are worth up to 229 basis points, and where, not coincidentally, the doom-loop multiplier also peaks.

3.2 Where stories spread

Under Assumption 1, the adoption probability is an increasing function of the price impact, so the epidemiology is a monotone transform of Proposition 1.

Proposition 2 (The susceptible zone). *(i) $R_0(\theta)$ is single-peaked in θ with vanishing tails. (ii) The susceptible zone $Z = \{\theta : R_0(\theta) \geq 1\}$ is therefore an interval $[\theta_L, \theta_H]$, nonempty if and only if amplification clears a threshold, $\bar{m}Ah(v_{\max}) \geq \gamma$. (iii) The zone widens with amplification A , with the story's bite ξ , with loss given default $1 - \mathcal{R}$, and with the feedback χ . It shrinks with the abandonment rate γ .*

Two features of the zone carry the paper's macro-history content. First, its existence is an era-dependent fact: at the calibration, cutting amplification to a quarter of baseline, a crude stand-in for the pre-social-media information environment, shrinks the zone to a set of width 0.001 in fundamentals units: effectively a point (Figure 2). The claim is not that stories were harmless before platforms. It is that supercritical *investor-to-investor* contagion at business-relevant speed plausibly requires modern amplification, and that the model makes the required threshold explicit. Second, the zone has a spread-space address: at baseline it maps to sovereigns trading between roughly 300 and 3394 basis points (Figure 1b), which is where the euro periphery lived from 2010 to 2012 and where narrative measurement should concentrate its power.

Corollary 1 (One statistic). *Price impact, contagion, and amplification are increasing transforms of a single local object, the belief-gap mass $\Phi(z^* + \Delta) - \Phi(z^*)$, itself proportional (for modest ξ) to the density $\phi(z^*)$ that governs the sensitivity of spreads to fundamentals news. A country-time measure of that sensitivity, the local slope of spreads in news, is therefore a sufficient statistic for boundary proximity: the model instructs empirical work to interact narrative measures with it, rather than with debt ratios or ratings directly.*

3.3 Three regions, and how a story kills

Combining the blocks, the state (θ_t, n_t) moves through a plane whose geometry the propositions have already drawn. Outside the zone, seeds die: prevalence decays at rate $\gamma(1 - R_0) > 0$, and the cumulative spread effect is bounded by the seed times the hump. Inside the zone, seeds grow logistically toward the endemic level $n^\infty = 1 - \gamma/\beta$, and spreads carry a narrative premium for as long as fundamentals remain febrile. Whether the premium kills depends on the feedback.

Proposition 3 (The moving cliff). *Let $D(\theta, n) = (1 - \rho_m)(\bar{\theta} - \theta) - \chi_m s^*(\theta, n)$ be the deterministic drift of fundamentals. (i) The no-story basin boundary $\hat{\theta}(0)$, below which drift is negative and the sovereign grinds to default absent shocks, rises with prevalence: $\hat{\theta}(n)$ is increasing, because s^* is increasing in n . (ii) There is therefore a critical interval $(\hat{\theta}(0), \hat{\theta}(n^\infty))$ of initial fundamentals*

at which the sovereign survives without the story and defaults with it, whenever the zone and the interval intersect. The intersection is nonempty at the calibration. (iii) Within the critical interval the crisis is slow: the outbreak must first build prevalence and then drain fundamentals, so default arrives with a lag governed by $\gamma(R_0 - 1)$ and χ_m , not by a jump.

Part (i) is the geometry worth remembering: prevalence does not push the sovereign toward a fixed cliff: it moves the cliff toward the sovereign. Part (iii) connects the mechanism to the crisis phenomenology it is meant to explain. Narrative crises in this model are not the instantaneous coordination failures of Cole and Kehoe (2000): they are months in which prevalence and spreads rise while fundamentals erode, visible in text data the whole way down: the slow-moving crises of Lorenzoni and Werning (2019), with an epidemic where their perfect foresight was. In the calibrated exhibit of Section 5, a sovereign seeded at two percent prevalence crosses the boundary at month 42. Its unseeded twin, facing identical fundamentals, survives indefinitely.

4 Policy: Deletion, Transparency, and the Cap

An intervention against a narrative can remove carriers, speed abandonment, or destroy the story's relevance. The three have different mathematics, and the differences are the policy content of the paper.

Proposition 4 (The counter-narrative trilemma). *(i) Deletion is postponement. A one-shot removal of a fraction f of carriers, with $R_0(\theta) > 1$ along the path, leaves the endemic outbreak unchanged and delays it by approximately $\ln(\frac{1}{1-f})/[\gamma(R_0 - 1)]$: at the calibrated zone center, deleting three quarters of carriers buys 1.5 months. (ii) Transparency is prophylaxis. Raising the abandonment rate γ , by making the story cheaper to check against verifiable data, scales R_0 down everywhere: it shrinks the zone from both ends, lowers the endemic level, and cuts the crisis wedge without any knowledge of which story will strike. (iii) A credible cap is sterilization. A facility that pegs the spread at $\bar{s}$ whenever the market would exceed it sets $\partial s^*/\partial n = 0$ wherever it binds. Under Assumption 1 adoption stops, R_0 falls below one, and the epidemic dies while the facility never needs to buy against a fundamentals-driven default it has not caused. The cap also severs the feedback χs , so it removes the critical region even for stories already endemic.*

Part (i) is the futility formula, and its empirical counterpart is the well-documented experience that rebuttals and takedowns do little against circulating financial rumors: against a supercritical story, removing carriers is bailing a boat with the leak still open. Part (iii) is the model's reading of July 2012. The speech, and the OMT design behind it, functioned as a conditional peg on the object the redenomination story needed to move. Once the story could not move prices, it stopped being worth telling, and its prevalence, not merely its price, collapsed. The model adds two qualifications that the episode's celebratory retellings omit. The sterilization claim is exactly as credible as the peg, and the required credibility is computable. If markets attach probability c to activation, the story's price relevance in capped states is $1 - c$ times its uncapped value, and with adoption proportional to relevance the reproduction number scales the same way:

sterilization requires $c \geq 1 - 1/R_0(\theta)$, which at the calibrated zone peak of $R_0 = 4$ means the facility must be believed with probability three quarters. A peg doubted more than that leaves the epidemic alive in the doubted states, its wedge the uncapped wedge scaled by the surviving relevance. And the instrument is time-sensitive because the feedback runs while institutions deliberate: the quantitative section prices delay at 17 percentage points of two-year default probability per year.

The disagreement between adoption rules, flagged at Assumption 1, becomes observable here. If adoption tracks price relevance, a credible cap kills narrative *volume*: text measures should collapse after the announcement. If adoption tracks perceived mispricing, the story survives unpriced: volume persists while spreads sit at the cap. The distinction is measurable in post-OMT text, and Section 6 registers it as a prediction rather than adjudicating it by assumption here.

5 Quantitative Magnitudes

5.1 Calibration

Table 1 lists parameters. The fundamentals block sets annual persistence 0.90 and innovation 0.05, so the unconditional standard deviation of capacity is about 0.11 against a long-run level of 0.15: default, at the boundary $\theta = 0$, is a genuine tail event for a healthy sovereign and a live one inside the zone. Recovery of 0.60 matches mean haircuts near 40 percent (Cruces and Trebesch, 2013). The feedback χ is not asserted, it is an identity. Capacity is measured in annual resources, so a year of elevated spreads costs the sovereign the spread times the share of debt refinanced within the year, which is the gross financing need: χ equals the GFN ratio. High-debt GFN runs from ten percent of output in moderate cases to thirty at crisis levels, Italy and Greece in 2011, so the baseline $\chi = 0.20$ is the representative high-debt point and $[0.10, 0.30]$ the examined band, with the headline numbers re-run at both ends below. The story's bite $\xi = 0.03$ shifts perceived capacity by roughly half an innovation: a serious accusation, not an apocalypse. On the epidemic side, abandonment $\gamma = 0.30$ per month gives an unforced story a three-month half-life, in the range of news-cycle persistence, and peak reproduction $R_0 = 4$ at baseline amplification is the model's amplification-era stand: Figure 2 shows the zone this opens and what remains of it at a quarter of the amplification. The seed, two percent of investor wealth, is a viral event, not a conspiracy.

5.2 The hump, the zone, and the outbreak

Figure 1 draws Proposition 1: the price relevance of the story peaks at 229 basis points per ten points of prevalence near the boundary and dies in both tails, and panel (b) places the susceptible zone in spread space, between 300 and 3394 basis points. Figure 2 draws Proposition 2: the baseline zone spans $[0.002, 0.070]$ in fundamentals. At a quarter of baseline amplification the zone degenerates to a point, and at 1.5 times baseline it reaches to $\theta = 0.081$.

Figure 3 is the mechanism end to end, deterministically. Three sovereigns are seeded with the same two percent prevalence. The safe one ($\theta_0 = 0.15$) sheds the story within months, and

Table 1: Calibration.

Parameter		Value	Category	Basis
Persistence (annual)	ρ	0.90	fixed	fundamentals half-life ≈ 6.6 years
Innovation sd (annual)	σ	0.05	internal	unconditional sd ≈ 0.11
Long-run capacity	$\bar{\theta}$	0.15	internal	safe-sovereign spread < 10 bp
Recovery	$\mathcal{R}$	0.60	estimated	haircuts $\approx 40\%$ (CT)
Feedback	χ	0.20	estimated	gross-financing-need arithmetic
Story bite	ξ	0.03	scenario	half an annual innovation, $\Delta = 0.54$
Abandonment	γ	0.30/mo	scenario	unforced story half-life ≈ 3 months
Peak reproduction	R_0	4	scenario	amplification era, pre-era = 1
Seed	n_0	0.02	scenario	a viral event
Loop gain (max)		0.64	implied	uniqueness margin, Lemma 1

Notes: CT is Cruces and Trebesch (2013). Categories: externally fixed, directly estimated, internally calibrated to a stated target, or scenario. The epidemic block (ξ, γ, R_0, n_0) is scenario throughout, pending the estimation of Design 4.

its spread never moves. The distressed one ($\theta_0 = 0.01$) defaults on fundamentals in four months, with and without the story: the narrative is redundant against a foregone conclusion. The zone sovereign ($\theta_0 = 0.06$, spread 465 basis points) is the exhibit: without the story it survives indefinitely, while with it, prevalence builds to 74 percent, the spread grinds from 465 toward four thousand basis points, an excess over its twin that reaches 3198 basis points, and capacity crosses zero at month 42. Forty-two months from tweet to default: the slowness is not a bug. It is the phenomenology of 2010–2012.

5.3 The crisis wedge and the policy scorecard

Figure 4 moves from exhibits to hazards: eight thousand simulated paths per starting point, with and without the seed, common shocks. The narrative wedge, the story’s addition to the 24-month default probability, is hump-shaped in initial fundamentals: at most 5.2 percentage points, at $\theta_0 = 0.04$, essentially zero for safe sovereigns (0.06 points at $\theta_0 = 0.15$) and vanishing near the boundary where both twins default. Panel (b) applies the policies: transparency (γ up by half) cuts the peak wedge to 3.0 points, and the 400 basis-point cap eliminates it (0.0), while separately relieving fundamentals default risk by 19.9 points through the severed feedback: the cap is both narrative policy and debt policy, and the decomposition reports the two honestly. Figure 5 prices timing at the wedge peak: a cap available in month one holds the hazard to 46.3 percent, and delayed to month thirteen, 63.4. The no-narrative benchmark is 66.1 and no intervention 71.3. Delay costs 17 points per year. One-shot deletion of three quarters of carriers, at its best timing, saves 2.1 points: the futility formula, drawn.

Since χ carries a data range rather than a point, the headline numbers are bands. Across the GFN band $[0.10, 0.30]$, with identical shocks and the epidemic block held at its calibrated amplification, the peak wedge runs from 0.7 to 6.9 percentage points, the annual cost of cap delay from 4 to 41 points, and the peak price relevance from 124 to 1457 basis points per ten points of prevalence. The zone survives at both ends, the wedge stays hump-shaped, and an earlier cap is better everywhere, so nothing qualitative hangs on the point estimate. Two ends

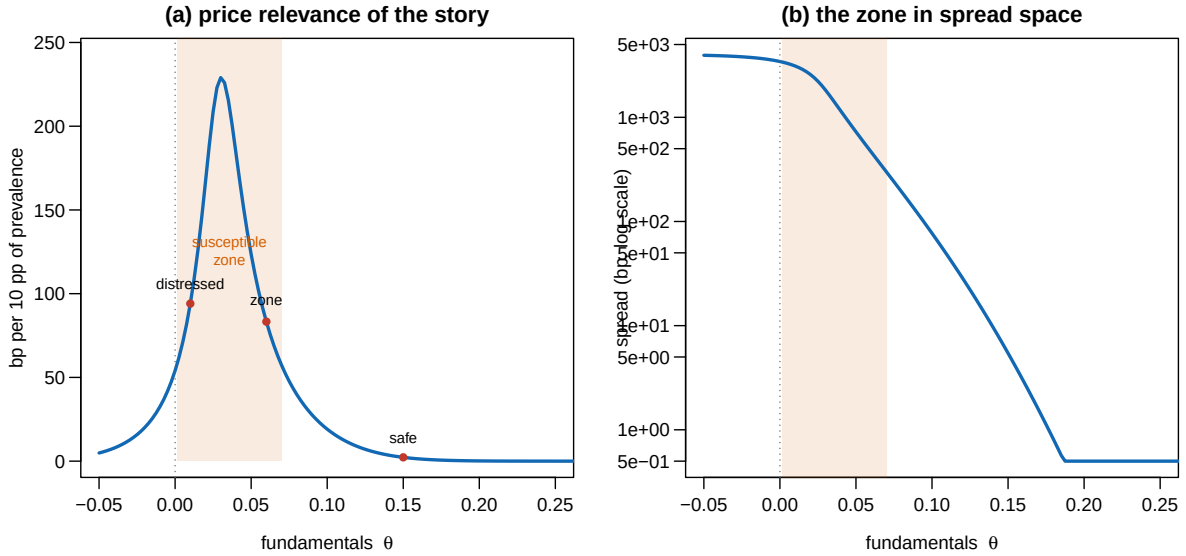


Figure 1: The derived hump. (a) Equilibrium price impact of the story, basis points per ten percentage points of prevalence. Shading marks the susceptible zone, and dots mark the three scenarios of Figure 3. (b) The equilibrium spread schedule: the zone in spread space.

of the band deserve their own sentences. At $\chi = 0.10$ the loop is mild, the deterministic zone sovereign never defaults within the simulation horizon, and the story is a tax rather than a killer. At $\chi = 0.30$ the maximal loop gain reaches 0.96 against the multiplicity bound of one: crisis-level financing needs put the pricing equilibrium at the edge of the Calvo (1988) region, the deterministic default arrives at month twelve rather than 42, and the attribution discipline that uniqueness buys is nearly exhausted. The model’s own arithmetic says that a sovereign with thirty percent gross financing needs is one where narrative effects and equilibrium selection stop being separable, which is a scope statement, not a footnote.

6 Measurement and Empirical Designs

6.1 Measuring an epidemic in text

The model’s belief object is not sentiment. It is a specific, portable story with a prevalence and a reproduction rate, and both are estimable. The measurement protocol, specified fully in Appendix E, has four layers. First, the story universe: five sovereign-debt narrative categories, default and restructuring, bailout and IMF rescue, redenomination and exit, fiscal-sustainability collapse, and the bank-sovereign loop, defined by claim structure rather than keyword lists. Second, classification: country-day text from newswires, national press, and platform data where licensed, classified by a transparent dictionary layer plus a supervised language-model layer whose prompts, labels, and validation statistics are all reported. Gentzkow et al. (2019) supply the standards, and the policy-uncertainty index of Baker et al. (2016) the precedent for index construction from counts. Third, the epidemiological quantities: prevalence $\hat{n}_{i,t}$ as the

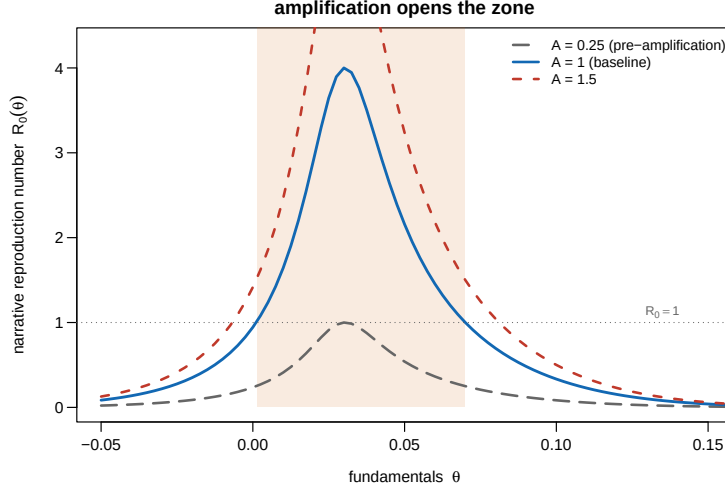


Figure 2: The narrative reproduction number $R_0(\theta)$ at three amplification levels. The susceptible zone is the superlevel set above one. At $A = 0.25$ it degenerates to a point.

reach-weighted share of country- i financial coverage carrying the story, and reproduction $\hat{R}_{i,t}$ from the cascade structure, the rate at which the story jumps outlets and platforms, in the spirit of the news-structure decompositions of Bybee et al. (2020). Fourth, validation: hand-labeled samples with inter-rater statistics, and the requirement, from Andre et al. (2026), that text-based prevalence predict survey beliefs, not only prices.

One definitional gap deserves its own sentence. The model’s n is the share of investor wealth pricing under the story, and no text archive measures that. What text delivers is the coverage share, and the protocol maintains, rather than proves, that coverage tracks the wealth-weighted object. The survey-validation layer is the check on that maintained link, not a formality: if reach-weighted coverage fails to predict the beliefs of surveyed investors, the prevalence series does not measure what the theory prices, and every design downstream inherits the failure.

6.2 Four designs

Design 1: the boundary interaction. The workhorse is a state-dependent local projection,

$$\Delta s_{i,t+h} = \alpha_i + \tau_t + \beta_h \hat{n}_{i,t} + \delta_h \hat{n}_{i,t} \times B_{i,t-1} + \Gamma'_h X_{i,t-1} + u_{i,t+h}, \quad (6)$$

with $B_{i,t}$ the sufficient statistic of Corollary 1: the rolling sensitivity of country i ’s spread to common fundamentals news, estimated from scheduled-release windows, not a debt ratio or rating. The model predicts $\beta_h \approx 0$ and $\delta_h > 0$: prevalence prices only where the hump is high, with the profile of δ_h across B bins tracing Figure 1a. Global factors in the sense of Longstaff et al. (2011) enter X and τ_t . The identification claim is not that text is exogenous but that the *state-dependence* of its pricing is the theory’s signature, which reverse causality running uniformly from spreads to coverage does not produce.

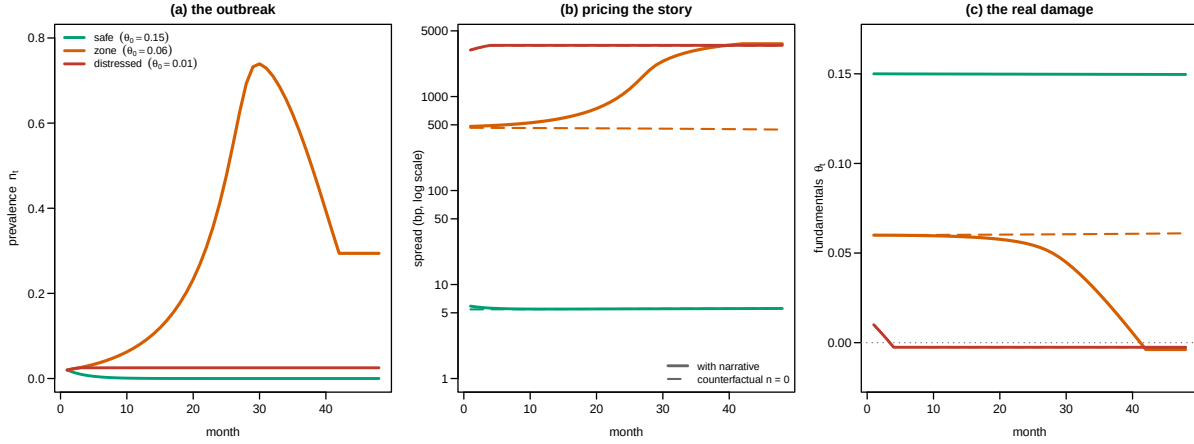


Figure 3: One seed, three sovereigns: deterministic paths from a two percent seed. (a) prevalence, (b) spreads against the $n \equiv 0$ counterfactual (dashed), (c) fundamentals. The zone sovereign defaults at month 42. Its counterfactual survives.

Design 2: scheduled releases, surprise against take-up. Scheduled fiscal releases separate information from its narrative echo: the surprise is the released number against consensus, and the take-up is the narrative prevalence response in the following 48 hours, orthogonalized on the surprise. The model predicts the take-up premium, the spread response to take-up beyond the surprise’s direct effect, exists inside the zone and not outside, and that identical surprises generate larger take-up where R_0 is higher: contagion, not content, does the marginal work.

Design 3: imported stories. Stories cross borders faster than fundamentals do. A crisis in country j raises the global prevalence of the default story, and its arrival in country i ’s discourse is predicted by pre-determined media, language, and investor-overlap exposure, not by i ’s own same-week fundamentals. Exposure-weighted foreign story shocks therefore instrument domestic prevalence, with real linkages, trade, bank claims, common creditors, controlled directly. I regard the exclusion restriction as defensible but not bulletproof, and the design as complementary: its value is that the model makes a sharp prediction about *where* imported stories price, the same hump, which a violation of exclusion through omitted fundamentals linkages would not reproduce. Platform outages and algorithm changes, transmission shocks in the proposal’s sense, are retained as auxiliary variation with the same caveat stated honestly.

Design 4: structural estimation and surveillance. Simulated method of moments on (ξ, χ, A, γ) , with moments chosen for state-dependence: the LP coefficients by boundary bin, outbreak persistence and final-size statistics, the text-spread lead-lag profile, and one event moment, the speed of prevalence collapse after the OMT announcements, which doubles as the test separating relevance-based from mispricing-based adoption (Proposition 4). The estimated model then runs forward as surveillance: country-time estimates $\hat{R}_{0,i,t}$, an epidemiological early-warning object with a structural interpretation, validated against the 2010–2012 chronology it was not fitted to.

Table 2 collects the tests. P2 deserves emphasis because it is the discipline the narrative

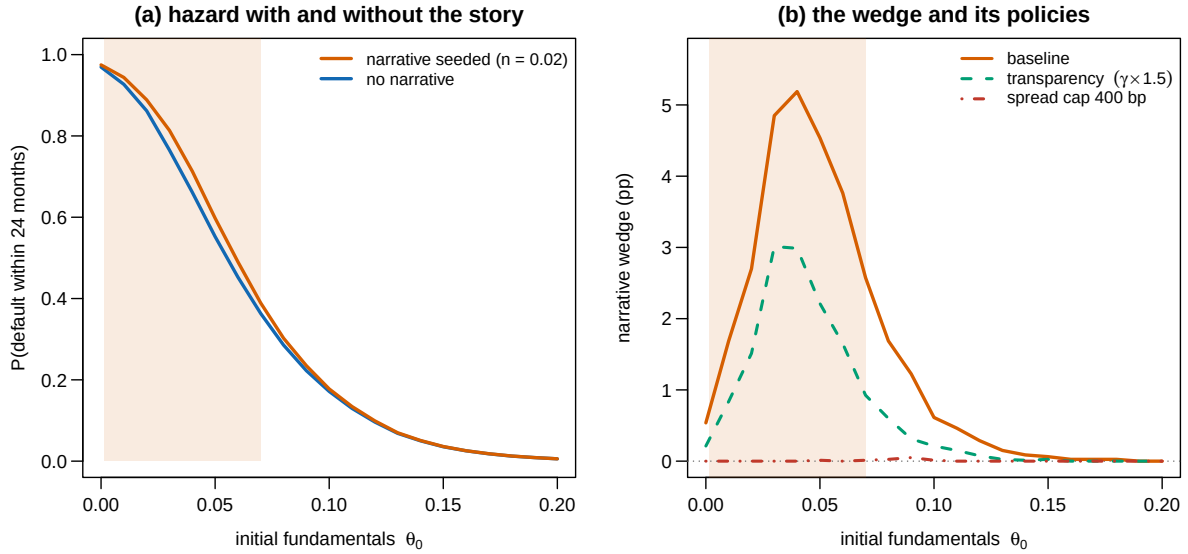


Figure 4: The narrative crisis wedge. (a) 24-month default probability by initial fundamentals, seeded against unseeded, common shocks. (b) The wedge under baseline, transparency, and a 400 bp cap (cap wedge measured against its own no-narrative baseline).

literature is usually accused of lacking: the theory supplies its own placebo. Stories go viral about safe sovereigns too, and the model requires those outbreaks to be loud in text and silent in prices. P4 separates the behavioral core from its rival with a single historical event’s text archive. And P6 embraces the reverse-causality critique rather than evading it: the model itself predicts that spreads lead text outside the zone, where stories are commentary, and text leads spreads inside it, where stories are the marginal pricing factor: the flip, not either direction alone, is the evidence.

Two of the six tests are decisive, and I state what their failure kills. If the Design-1 interaction δ_h is zero with confidence excluding the hump’s calibrated peak, or if prevalence prices safe sovereigns as strongly as zone sovereigns, the boundary structure is wrong and the paper’s discipline collapses back into the everything-explaining narrative residual it was built to escape. If narrative volume survives a credible cap at its pre-announcement level, Assumption 1 fails in favor of its mispricing rival, and the sterilization result, the paper’s sharpest policy claim, goes with it. The remaining tests calibrate rather than kill: their failure would move magnitudes, not the mechanism.

One validation moment already exists and entered no calibration. The fundamentals-unexplained spread components that De Grauwe and Ji (2013) document for the euro periphery, several hundred basis points at the 2011 peak, sit inside the model’s outbreak range: the calibrated excess spread reaches 3198 basis points at the top of an outbreak, and nothing in Table 1 was chosen to match their estimates. The 2010–2012 chronology plays the same role for Design 4: the surveillance model is to be validated against it, not fitted to it.

Table 3 documents feasibility. The text pipeline is the project’s main data investment:

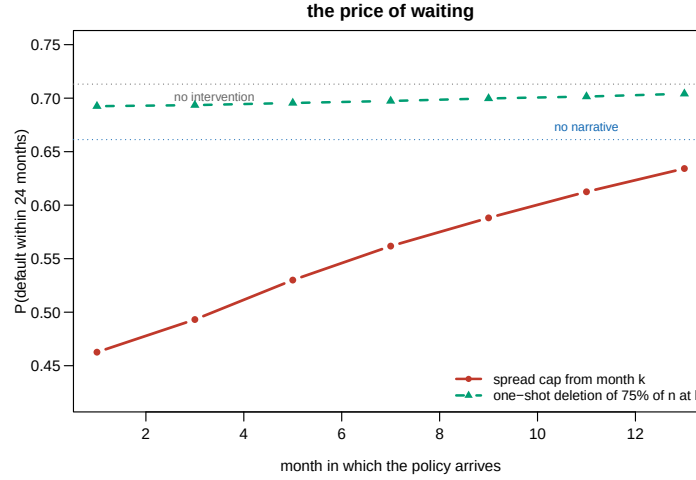


Figure 5: The price of waiting, at the wedge peak: 24-month default probability as a function of the month in which the policy arrives.

Table 2: Predictions and status.

Prediction	Model tude	magni- tude	Null	Status
P1: prevalence prices spreads only near the boundary, profile of δ_h traces the hump	229 vs 2.3 bp per 10 pp		uniform pricing	pending data
P2: outbreaks in immune countries move attention, not spreads (built-in placebo)	wedge 0.06 pp		spreads move	pending data
P3: identical fiscal surprises generate larger narrative take-up inside the zone	R_0 : 4 vs < 1		uniform take-up	pending data
P4: a credible cap collapses narrative volume, not only its price	Prop. 4(iii)		volume persists	pending data
P5: transparency reforms cut the pricing of narrative shocks	wedge $5.2 \rightarrow 3.0$ pp		no change	pending data
P6: text leads spreads inside the zone, spreads lead text outside	lead-lag flip		one direction	pending data

newswire archives are licensed and platform text is both licensed and, for the historical window, incompletely archived, which is why the protocol leans on newswires for the 2010–2012 core. Everything on the pricing side is standard.

7 Conclusion

The narrative hypothesis in macroeconomics has suffered from explaining too much. This paper’s contribution is a boundary: a model in which stories about sovereign default are priced and spread, and occasionally kill, but only in a derived interval of fundamentals, and in which every step of that sentence is measurable. One primitive, the S-curve of default probability, generates the pricing hump, the contagion zone, and the doom-loop amplification as three faces of the same

Table 3: Data feasibility.

Dataset	Coverage	Frequency	Access	Observed or proxied	Status
Newswire and press text	Factiva/LexisNexis, 1995–	daily	licensed	prevalence proxied by coverage	pending license
GDELT event text	global, 2015–	daily	public	coverage counts	in hand
Platform text	where licensed, post-2010	real time	restricted	coverage, partial archive	to assemble
Consensus forecasts	major sovereigns	monthly	licensed	observed	pending license
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	pending license
Fiscal-release calendars	national and Euro-stat	event	public	observed	in hand
Survey crash beliefs	Andre et al. (2026)-type	occasional	public	observed	in hand
Media-reach weights	circulation, audience data	annual	public	observed	to assemble

local statistic: where beliefs are priced, stories about beliefs are worth telling. The equilibrium is unique throughout: no sunspots are selected, and the belief object is an epidemic with an estimable prevalence and reproduction number rather than a residual with a name.

The quantitative lessons are asymmetric and, I think, credible for being so. Stories tax vulnerable sovereigns heavily, up to 229 basis points per ten points of prevalence, thousands at outbreak peak, and change outcomes rarely: at most 5.2 percentage points of two-year default hazard, in a narrow band. The policy lessons rank interventions by their epidemiology: deletion postpones, transparency inoculates, and a credible cap sterilizes, at a price of 17 percentage points per year of delay. The whatever-it-takes speech, on this reading, was not communication policy: it was the destruction of a story’s reason to exist.

What the paper does not yet have is the data its empirical half specifies: the classified story archive, the prevalence and reproduction series, and the estimated (ξ, χ, A, γ) that would convert the calibrated magnitudes into measurements. The measurement protocol is deliberately the most detailed part of the empirical agenda, because the project lives or dies on whether narrative prevalence can be measured as transparently as a price. If it can, the payoff extends beyond this paper: an epidemiological surveillance system for sovereign debt markets, tracking not whether investors are gloomy but which stories are spreading, how fast, and whether the sovereign beneath them is standing where stories kill.

A Proofs

A.1 Proof of Lemma 1

Write $F(s) = (1-\mathcal{R})\bar{p}(\theta, s, n)$. Then $F'(s) = (1-\mathcal{R})\frac{\chi}{\sigma}[(1-n)\phi(z)+n\phi(z+\Delta)] \leq (1-\mathcal{R})\frac{\chi}{\sigma}\phi(0) < 1$ under the stated condition, so F maps $[0, 1-\mathcal{R}]$ into itself with slope uniformly below one: a contraction, with unique fixed point s^* . Monotonicity in (θ, n) follows from $\bar{p}$ increasing in n (since $\Phi(z+\Delta) > \Phi(z)$) and decreasing in θ , plus the contraction property. The multiplier is $1/(1-F'(s^*))$ by the implicit function theorem. If instead $(1-\mathcal{R})\chi\phi(0)/\sigma > 1$, F is S-shaped with slope exceeding one near $z^* = 0$, and an interval of θ admits three fixed points: Appendix B. A cap replaces F by $\min(F, \bar{s})$, preserving the contraction. ■

A.2 Proof of Proposition 1

The gap $G(z) = \Phi(z+\Delta) - \Phi(z) = \int_z^{z+\Delta} \phi(u) du$ is the convolution of the log-concave density ϕ with the indicator of $[-\Delta, 0]$, hence log-concave by the Prékopa–Leindler theorem. $G'(z) = \phi(z+\Delta) - \phi(z)$ vanishes only at $z = -\Delta/2$, so G is single-peaked there, where $\Phi(z^* + \Delta/2) \approx \bar{p} \approx 1/2$ up to the window's curvature, and $G \rightarrow 0$ in both tails. Monotonicity of $z^*(\theta)$: differentiating the fixed point, $dz^*/d\theta = -(\rho/\sigma)M < 0$. For (iii), $1 \leq M \leq \bar{M} = [1 - (1-\mathcal{R})\chi\phi(0)/\sigma]^{-1}$, and $M \rightarrow 1$ in both tails since $\phi \rightarrow 0$, so the equilibrium impact $(1-\mathcal{R})GM$ vanishes in both tails and lies within the factor $\bar{M}$ of the single-peaked G . Single-peakedness of the product itself holds under the mild condition $\Delta \leq 2$: on $z \leq -\Delta/2$ both G and M are increasing (the latter because ϕ is), on $z \geq 0$ both are decreasing, and on $[-\Delta/2, 0] \subset (-1, 1)$ both are log-concave, M because $\frac{d^2}{dz^2} \ln M$ has the sign of $(z^2 - 1)(1 - \kappa\phi) + \kappa\phi z^2 < 0$ there (with $\kappa = (1-\mathcal{R})\chi/\sigma$), so their product is log-concave on the bridge and the peak is unique. ■

A.3 Proof of Proposition 2

$R_0 = \bar{m}A h(v)/\gamma$ with h increasing and v single-peaked with vanishing tails (Proposition 1): R_0 is single-peaked with vanishing tails, so its superlevel set at one is an interval, nonempty iff the peak clears one, which is the stated amplification threshold. Comparative statics: A scales R_0 , ξ widens the window defining G , $1-\mathcal{R}$ and χ scale v through the pricing equation and the multiplier, and γ divides. Each moves R_0 monotonically at every θ , so the superlevel interval moves as stated. ■

A.4 Proof of Proposition 3

(i) $D(\theta, n)$ is strictly decreasing in n because s^* is strictly increasing in n (Lemma 1) wherever $G > 0$. Define $\hat{\theta}(n) = \sup\{\theta : D(\theta, n) < 0\}$: since the set $\{D < 0\}$ expands with n , $\hat{\theta}$ is nondecreasing, strictly where the pricing hump is active. (ii) For $\theta_0 > \hat{\theta}(0)$ the unseeded deterministic path has $D > 0$ throughout and converges upward toward $\bar{\theta}$: survival. For seeded paths with θ_0 in the zone, prevalence grows toward n^∞ , and once n_t is large enough that $\hat{\theta}(n_t) > \theta_t$, drift turns negative and both θ_t and, through the hump, R_0 move adversely: the pair

enters a trapping region and crosses the boundary in finite time. The set of θ_0 for which this occurs is nonempty whenever $(\hat{\theta}(0), \hat{\theta}(n^\infty))$ meets the zone. At the calibration $\theta_0 = 0.06$ exhibits it. (iii) The default time decomposes into the outbreak’s build time, of order $\ln(n^\dagger/n_0)/[\gamma(R_0 - 1)]$ for the prevalence $n^\dagger$ at which drift turns negative, plus the drain time $(\theta_0 - 0)/|\bar{D}|$: both are months-to-years for calibrated parameters, never a jump. ■

A.5 Proof of Proposition 4

(i) With $R_0 > 1$ roughly constant along the path, linearized prevalence grows as $n_t \approx n_0 e^{gt}$, $g = \gamma(R_0 - 1)$, and a cut to $(1 - f)n$ is undone when $e^{gt} = 1/(1 - f)$. The logistic’s later stages and final size depend only on β, γ : unchanged. (ii) γ enters R_0 as a divisor: the superlevel set of the single-peaked R_0 shrinks from both ends, and $n^\infty = 1 - \gamma/\beta$ falls at every interior θ . (iii) Where the cap binds, s^* is locally constant in n , so $v = 0$, $h(0) = 0$, $\beta = 0 < \gamma$: prevalence decays at rate γ regardless of R_0 ’s uncapped value. A cap at or below the zone’s entry spread binds throughout the zone. The calibrated 400 bp cap leaves a sliver of the zone’s upper edge unbound where R_0 barely exceeds one, and the measured crisis wedge there is zero (Figure 4b). Severing χs above the cap also raises D , shrinking $\{D < 0\}$ and hence the critical region, for any prevalence path. ■

B The Pricing Block: Details and the Multiplicity Frontier

Spread mapping. Linear expected-loss pricing $s = (1 - \mathcal{R})\bar{p}$ understates convexity at high $\bar{p}$. The exact zero-profit mapping $s = -\log(1 - \bar{p}(1 - \mathcal{R}))$ steepens the hump and widens the zone, so the text’s linear choice is conservative. Endogenizing the promised yield through an issuance requirement adds a second loop with the same sign.

The Calvo frontier. The uniqueness condition $(1 - \mathcal{R})\chi\phi(0)/\sigma < 1$ fails when the feedback is strong: at the calibration the maximal gain is 0.64, and at the top of the GFN band ($\chi = 0.30$) it reaches 0.96: crisis-level financing needs sit at the frontier’s edge without crossing it. Past one, an interval of fundamentals carries three pricing equilibria, the static multiplicity of Calvo (1988) studied quantitatively by Aguiar et al. (2022). The narrative mechanism neither requires nor excludes that regime: inside it, a story could additionally coordinate jumps between equilibria, a compounding this paper deliberately avoids by calibrating into the unique region, where every narrative effect is attributable and measurable.

Wealth-weighted pricing. Averaging beliefs by wealth presumes infected and susceptible investors hold with equal intensity. Letting the infected also shorten duration or exit produces an effective prevalence exceeding n in the pricing equation and strengthens every result. The text’s specification is again the conservative one.

C Simulation Details

Monthly parameters derive from annual ones by $\rho_m = \rho^{1/12}$, $\sigma_m = \sigma/\sqrt{12}$, $\chi_m = \chi/12$. The pricing fixed point is solved by damped iteration to 10^{-12} on a 226×51 grid over $(\theta, n) \in [-0.10, 0.35] \times [0, 1]$, bilinearly interpolated in simulation. Monte Carlo experiments use 8,000 paths of 24 months with common random numbers across configurations, seed fixed. Default is absorbing at $\theta < 0$. Deterministic exhibits shut the fundamentals shock and freeze paths at the boundary. The one-shot deletion experiment removes 75 percent of prevalence in the stated month, and the cap experiment switches the pricing table from uncapped to capped from the stated month onward. All figures and every number in the text are generated by the accompanying Julia and R code from a single JSON of results.

D Calibration Audit

Table 4 records where each parameter enters, its units, its source, the range examined, and what moves. The mechanism disappears at three nameable places. If amplification falls below the threshold of Proposition 2(ii), the zone is empty and every narrative is a fizzle: the pre-platform world. If $\chi = 0$, stories are priced but cannot kill: the critical region of Proposition 3 is empty and the wedge is zero at every θ_0 . And if the loop gain crosses one, the pricing equilibrium multiplies and the narrative effects stop being attributable: the calibration sits at 0.64 against that bound.

Table 4: Calibration audit.

Parameter	Enters at	Units	Source or moment	Range examined	Effect of the range
ρ, σ	eq. (2)	annual, capacity	half-life, unconditional sd	none	scale z and Δ jointly
$\bar{\theta} = 0.15$	eq. (1)	capacity	safe spread < 10 bp	none	anchors the safe tail
$\mathcal{R} = 0.60$	pricing	recovery rate	Cruces and Trebesch (2013)	none	scales s^* and the hump
$\chi = 0.20$	eqs. (1), (2)	capacity per bp-year	GFN identity, IMF range	0.10 to 0.30	wedge 0.7 to 6.9 pp, gain 0.96 at the top, $\chi = 0$ kills the cliff
$\xi = 0.03$	eq. (3)	capacity	scenario, half an innovation	via Δ	hump height and width
$\gamma = 0.30$	eq. (4)	per month	news-cycle half-life	$\times 1.5$ (transparency)	wedge 5.2 to 3.0 pp
$R_0 \text{ peak} = 4$	eq. (5)	unitless	scenario, amplification era	$A \in \{0.25, 1, 1.5\}$	zone width 0.001 to [0.002, 0.081]
$n_0 = 0.02$	initial condition	wealth share	scenario, event	timing experiments	delay formula only
cap = 400 bp	policy	spread	OMT-scale instrument	month 1 to 13	hazard 46.3 to 63.4 percent
loop gain	Lemma 1	unitless	implied	2χ crosses one	multiplicity, App. B

Notes: the fundamentals and payoff blocks are fixed or estimated, the epidemic block is scenario throughout, and Design 4 is the declared path to estimating (ξ, χ, A, γ) .

E Data and Measurement Protocol

Sources. Newswire and press text via Factiva/LexisNexis and GDELT, platform text where licensed, consensus forecasts for scheduled-release surprises, sovereign bond and CDS spreads at daily frequency, scheduled fiscal-release calendars, debt-management-office auction results, and media-reach weights from circulation and audience data.

Classification. Each document is classified into the five narrative categories by (i) a transparent dictionary layer requiring a sovereign-solvency claim structure, subject, verb class, and horizon, not keywords alone, and (ii) a supervised language-model layer trained on hand-labeled samples, with prompts, label definitions, training splits, and error matrices published in full. Inter-rater reliability on the hand-labeled core and out-of-sample precision/recall by category are reported alongside every index release.

Construction. Prevalence $\hat{n}_{i,t}$: reach-weighted share of country- i sovereign-finance coverage carrying the category, smoothed at the weekly frequency the SIR block implies. Reproduction $\hat{R}_{i,t}$: cascade growth across outlets and platforms within story clusters, estimated by the standard epidemiological generation-interval method applied to first-appearance timestamps. Validation: correlation with survey-based crash beliefs where available (Andre et al., 2026), and the P2 placebo, prevalence spikes for safe sovereigns must not price.

F Extensions

Cross-border story networks. Let prevalence in country i respond to a reach-weighted sum of foreign prevalences. Stories then comove across borders more than fundamentals do, generating correlated spread movements among zone countries that fundamentals-based contagion cannot explain: a candidate microfoundation for the common factors of Longstaff et al. (2011), and the structure behind Design 3.

Endogenous amplification. Attention to sovereign stories rises with volatility, making A procyclical: the zone widens exactly when fundamentals deteriorate, an accelerant with the same signature as a rising multiplier and a natural explanation for why crisis eras feel narrative-dense.

Immunity and waves. Investors who abandoned a story once may resist it: an SIRS structure with slow waning of immunity produces refractory periods after outbreaks and recurring waves at business-cycle frequency, testable in long text archives: the same story striking the same country twice should spread slower the second time, controlling for the boundary.

Heterogeneous stories. A mixture over bite sizes ξ concentrates the priced composition toward severe stories as the boundary nears, since the hump’s window widens with Δ : near the

boundary, discourse should not only grow but darken, a compositional prediction the classification layer can test.

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